

Last class

X, Y NLS

$T: X \rightarrow Y$ (linear transformation)

Then

- 1) T is cont at $0 \in X$
- 2) T is cont on X
- 3) T is unif. cont

4) $\exists c > 0$ s.t.
 $\|T(x)\| \leq c \|x\|, \forall x \in X$ $\rightarrow$ bddness

5) $\{T(x) : x \in X, \|x\|=1\}$ is a bdd subset of Y .

6) For every bdd $E \subseteq X$, $T(E)$ is bdd.

equivalent.

$(3) \Rightarrow 2 \Rightarrow 1$ obvious

$4 \Rightarrow 1$

Proved

$4 \Rightarrow 6 \Rightarrow 5$

Today $(3) \Rightarrow 2 \Rightarrow 1$ ~~Proved~~
 $4 \Rightarrow 1$ ~~Proved~~ Today $4 \Rightarrow 6 \Rightarrow 5$

To prove $(1) \Rightarrow (3)$

Let $\epsilon > 0$

Since T is continuous at $0 \in X \quad \exists \delta > 0$
s.t

$$\|x - 0\| < \delta$$

$$\Rightarrow \|T(x) - T(0)\| < \epsilon$$

ie,

$$\|x\| < \delta \Rightarrow \|T(x)\| < \epsilon \quad \forall x \in X.$$

$\forall u \in X$

$$x - u \in X.$$

$$\therefore \|x - u\| < \delta \Rightarrow \|T(x - u)\| < \epsilon$$

$$\Rightarrow \|T(x) - T(u)\| < \epsilon \quad \forall x, u$$

T is unif. cont.

$$\underline{(5)} \Rightarrow (4).$$

Let $x \in X, \quad x \neq 0$

$$\tilde{x} = \frac{x}{\|x\|} \Rightarrow \|\tilde{x}\| = 1$$

$$\|T(x)\| \leq M \quad (\text{Given}).$$

$$\Rightarrow \|T(x)\| \leq M \|x\|$$

$$\Rightarrow T \text{ is } \underline{\text{bdd.}}$$

X, Y NLS

$[T: X \rightarrow Y \text{ linear transformation}$
 is continuous $\Leftrightarrow$ For every
 bdd set E in X
 $T(E)$ is bdd in $Y]$

T is continuous $\Rightarrow$ is also bdd.

Ex.

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$\|x\|_2 = \left(\sum_{j=1}^n |x(j)|^2 \right)^{1/2}$$

$$T(x(1), \dots, x(n)) \\ = (0, x(2), x(3), \dots, x(n))$$

- T is a linear transformation.
- $x \in \mathbb{R}^n$

$$\|T(x)\|_2^2 = \sum_{j=2}^n |x(j)|^2$$

$$= \|x\|_2^2 - |x(1)|^2$$

$$< \|x\|_2^2$$

$$\boxed{\|T(x)\| \leq c \|x\|}$$

any norm
in $\mathbb{R}^n$.

Theo. X be a finite dim. NLS.
 Y be a NLS.

Then every linear transformation
from $X \rightarrow Y$ is bdd (cont).

Pf. $X = \{0\}$ N.T.P.

Let $X \neq \{0\}$

$\exists \{u_1, \dots, u_m\}$ basis of X .

Let (x_n) be a seq in X and
 $x_n \rightarrow \underline{x}$ in X .

$$x_n = k_{n,1}u_1 + k_{n,2}u_2 + \dots + k_{n,m}u_m$$

$k_{n,j} \in \mathbb{K}.$

We have

$$x = k_1 u_1 + k_2 u_2 + \dots + k_m u_m$$

Claim $x_n \rightarrow x \Rightarrow k_{n,j} \rightarrow k_j$ as $n \rightarrow \infty$. ✓

$$\|x_n - x\| = \|(k_{n,j} - k_j)u_j + \underbrace{\sum_{\substack{n=1 \\ n \neq j}}^m (k_{n,n} - k_n)u_n}\|$$

$\in Y_j$

(Define $Y_j = \text{span} \{u_i, \substack{i=1, \dots, m \\ i \neq j}\}$)

$\geq \text{dist}((k_{n,j} - k_j)u_j, Y_j)$

$$= |k_{n,j} - k_j| \text{dist}(u_j, Y_j)$$

Now $u_j \notin Y_j = \overline{Y_j}$ ($\because Y_j$ is finite dim).

$$\therefore \text{dist}(u_j, Y_j) > 0.$$

$$|k_n - k_j| \leq \frac{\|x_n - x\|}{\text{dist}(u_j, Y_j)} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\begin{aligned} T(x_n) &= T(k_{n,1} u_1 + \dots + k_{n,m} u_m) \\ &= k_{n,1} T(u_1) + \dots + k_{n,m} T(u_m) \\ &\rightarrow k_1 T(u_1) + \dots + k_m T(u_m) \\ &\Rightarrow = T(k_1 u_1 + \dots + k_m u_m) \\ &= T(x). \end{aligned}$$

$\therefore T$ is cont.

Ex.

$$X = C_{\infty}, \quad \|\cdot\|,$$

$$f: X \rightarrow \underline{\mathbb{K}}.$$

$$f(x) = f(x(1), x(2), \dots)$$

$$= \sum_{j=1}^{\infty} x(j)$$

$\therefore$ Clearly f is linear transformation.

$$|f(x)| \leq \sum_{j=1}^{\infty} |x(j)| = \|x\|,$$

$\therefore f$ is bdd.

$$(c_0, \|\cdot\|_2)$$

f is not bdd.

$$x_n = (1, \frac{1}{2}, \dots, \frac{1}{n}, 0, \dots, 0)$$

$$\|x_n\|_2^2 = \sum_{j=1}^n \frac{1}{j^2} \leq \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6}.$$

$$f(x_n) = 1 + \frac{1}{2} + \dots + \frac{1}{n} + 0 + \dots$$

$$= \sum_{j=1}^n \frac{1}{j}$$

~~So~~ Suppose f is bdd.

ie $\exists M > 0$ s.t

$$|f(x)| \leq M \|x\|_2$$

$$\sum_{j=1}^n \frac{1}{j} \leq \frac{M \pi^2}{6}. \quad \Downarrow$$

$\therefore$ Such M does not exist.

Theorem

Let X, Y be NLS & $Y \neq \{0\}$.

If X is infinite dimensional
then $\exists$ an unbdd lin. trans.
from $X \rightarrow Y$.

$\square$

$$T(u_n) = nb, \quad u_n \in E$$

$$= 0 \quad \text{o.w.}$$

$\Rightarrow$

$$T(x_n) = T\left(\frac{1}{n} \frac{u_n}{\|u_n\|}\right)$$

$$= \frac{1}{n \|u_n\|} \quad n.b$$

$$= \frac{b}{\|u_n\|} \xrightarrow[n \rightarrow \infty]{} 0$$

$\therefore T$ is not continuous.

Ex. $X = C^1[0,1] \rightarrow$ cont diff once.

$$Y = C[0,1]$$

$\rightarrow \left\{ f: [0,1] \rightarrow \mathbb{R} \mid f \text{ is } \cancel{\text{cont}} \text{ diff} \right.$
 $\left. \& \underline{f'} \text{ is cont} \right\},$

$$(X, \|\cdot\|_\infty) \quad (Y, \|\cdot\|_\infty)$$

$T(x) = \underline{x'}$ $\rightarrow$ differentiation
op.

T is linear.

Claim T is unbdd.

Consider

Theo. Let X, Y be NLS.

$T: X \rightarrow Y$ linear map.

T is bdd $\Leftrightarrow \underline{Z(T)}$, the
null space of T is
closed.

$$\textcircled{\tilde{T}}: X | Z(T) \rightarrow Y$$

$$\tilde{T}(x + Z(T)) = T(x)$$

is cont.

Proof: T is bdd. $X_0 \subseteq X$
 $\|x + x_0\| = \text{dist}(x, X_0)$ X/X_0
 X_0 closed

$$\begin{aligned} Z(T) &= \{x \in X \mid T(x) = 0\} \\ &= T^{-1} \{0\} \end{aligned}$$

Since T is bdd or continuous
so $Z(T)$ is closed.

Define. $\tilde{T}: X/Z(T) \rightarrow Y$

$$\tilde{T}(x + z(\tau)) = T(x).$$

$\tilde{T}$ is well defined. ie

$$x + z(\tau) = y + z(\tau)$$

$$\Rightarrow x - y \in z(\tau)$$

$$\Rightarrow T(x - y) = 0 \Rightarrow T(x) = T(y)$$

$$\Rightarrow \tilde{T}(x + z(\tau)) = \tilde{T}(y + z(\tau))$$

Also $\tilde{T}$ is linear.

Show

$$\|\tilde{T}(x + z(\tau))\| \leq M \|x + z(\tau)\|.$$

Choose $u \in Z(\tau)$

$$\| \tilde{T}(x + Z(\tau)) \|$$

$$= \| \tilde{T}(x + u + Z(\tau)) \|$$

$$= \| \underline{T}(x + u) \|$$

$$\leq \inf_{u \in Z(\tau)} \|x + u\| \quad \forall u \in Z(\tau)$$

$$= M \|x + Z(\tau)\|$$

$\therefore \tilde{T}$ is bdd.

To prove T is bdd.

$$\begin{aligned}
\|T(x)\| &= \|\tilde{T}(x + Z(\tau))\| \\
&\leq M \|x + Z(\tau)\| \\
&= M \inf \left\{ \|x - u\| \mid u \in Z(\tau) \right\}.
\end{aligned}$$

In particular

$$\begin{aligned}
\|T(x)\| &\leq M \|x\| \quad (\because 0 \in Z(\tau)) \\
\therefore T &\text{ is bdd.}
\end{aligned}$$

Let X, Y be NLS.

$L(X, Y)$ = space of all
linear transformations

$B(X, Y)$ = space of all bdd
linear transformation.

- $B(X, Y) \subseteq L(X, Y)$

- X is finite dim.

$$B(X, Y) = L(X, Y)$$

- X is infinite dim

$$B(X, Y) \subsetneq L(X, Y).$$

$$(T_1 + T_2)(x) = T_1(x) + T_2(x)$$

$$(\alpha T)(x) = \alpha T(x)$$

$$\| (T_1 + T_2)(x) \| \leq M (\|x\|) .$$

(Triangle inequality) .

Qn. Can we define a norm
on $B(X, Y)$?

If $T \in B(X, Y)$.

$$\|T(x)\| \leq M \|x\| . \quad \forall x \in X .$$

$$\frac{\|T(x)\|}{\|x\|} \leq M . \quad \forall x \in X, \quad x \neq 0$$

$$\sup \left\{ \frac{\|T(x)\|}{\|x\|} \mid x \in X \right\} < \infty .$$

Define

$$\|T\| = \sup \left\{ \frac{\|T(x)\|}{\|x\|} \mid x \in X, x \neq 0 \right\}.$$

• $T \in B(X, Y)$.

$$\frac{\|T(x)\|}{\|x\|} \leq \|T\| \quad \forall x \in X.$$

$$\Rightarrow \|T(x)\| \leq \|T\| \|x\|.$$